

U-Net with SSM-inspired Gated Large-Receptive-Field Refinement for fetal abdomen Segmentation and Circumference Measurement using Ultrasound Images

Vrushti Patel, Jayendra Bhalodiya
School of Engineering and Applied Science
Ahmedabad University,
Ahmedabad, India

vrushti.p@ahduni.edu.in, jayendra.bhalodiya@ahduni.edu.in

Abstract—Ultrasound images are widely used to monitor fetal growth, but automatic segmentation of fetal structures is difficult due to noise, motion artifacts, and image variability. This study addresses these challenges using deep neural networks. This study contributes a U-shaped encoder–decoder model with state-space model (SSM)-inspired large-receptive-field refinement blocks for fetal abdomen segmentation and fetal abdomen circumference measurement. Experiments are conducted using the benchmark dataset, ACOUSLIC-AI [1], and by following five-fold cross validation and ten-fold cross validation strategies. Standard measures of Dice coefficient were used to evaluate performance in segmentation and the accuracy of the estimated abdominal circumference was evaluated using the Normalized Absolute Error (NAE-AC).

Index Terms—Medical image segmentation, fetal abdomen ultrasound, U-Net, state-space models (SSM), Vision Mamba, deep learning, blind-sweep ultrasound imaging.

I. INTRODUCTION

Monitoring fetal growth during pregnancy is an important component of prenatal care, as abnormal growth patterns are associated with increased risks of neonatal morbidity, mortality, and long-term health complications [4]–[6]. In clinical practice, fetal biometric measurements obtained from ultrasound imaging are commonly used to assess fetal development and identify potential growth abnormalities [9], [12]. Fetal abdominal circumference is a common measure for fetal growth and fetal nutrition status [4]. Manual analysis of ultrasound images is however subjective, needs skilled operators, and may be influenced by the fluctuation of image quality and the conditions under which images are taken.

Blind-sweep ultrasound acquisition protocols are used to improve access to prenatal imaging in low-resource settings, where access to trained sonographers and conventional image analysis infrastructure may be limited [14], [15]. The ACOUSLIC-AI challenge dataset highlights such challenges, providing blind-sweep ultrasound data aimed at automated fetal abdominal circumference measurement [1]. Such data contains significant speckle noise, motion artifacts, and irregular anatomical planes, making automatic segmentation

challenging. In this context, point-of-care ultrasound (POCUS) offers a practical route for prenatal screening because it can be performed with portable devices by minimally trained operators, but it also increases variability in acquisition and makes it harder to consistently capture standard anatomical planes needed for accurate biometric assessment. This makes robust automated methods particularly important for supporting clinical decision-making in resource-constrained environments.

Deep learning-based approaches have been increasingly explored for medical image analysis tasks, including ultrasound-based fetal biometric estimation [16]–[18], [20]. Encoder–decoder architectures such as U-Net are commonly used for segmentation due to their ability to capture hierarchical image features while preserving spatial details through skip connections [2]. In parallel, recent work in computer vision has investigated SSM based architectures that model broader contextual interactions across spatial locations [3].

In this study, we investigate the integration of state space model-inspired large receptive-field refinement blocks to the U-shaped encoder-decoder segmentation framework to segment the fetal abdomen in ultrasound images. The pipeline used in the experiment consists of preprocessing, dataset preparation, implementing the model, and training and evaluating the model using standard segmentation measures. In addition to segmentation performance, the predicted fetal abdomen masks were used to estimate abdominal circumference, and the accuracy of the measurement was evaluated using the NAE-AC. This work is therefore aligned with POCUS-AI efforts aimed at enabling scalable, operator-independent prenatal screening from blind-sweep ultrasound data.

II. METHODOLOGY

A. Overview

The model adapts a U-shaped encoder-decoder structure where there are four encoder stages, which are connected to a bottleneck and the four decoder stages, which are symmetrical. Two experimental approaches were performed, five-fold cross

validation and ten-fold cross validation which allows generalizability assessment of the model. The entire model pipeline is indicated in the figure below:

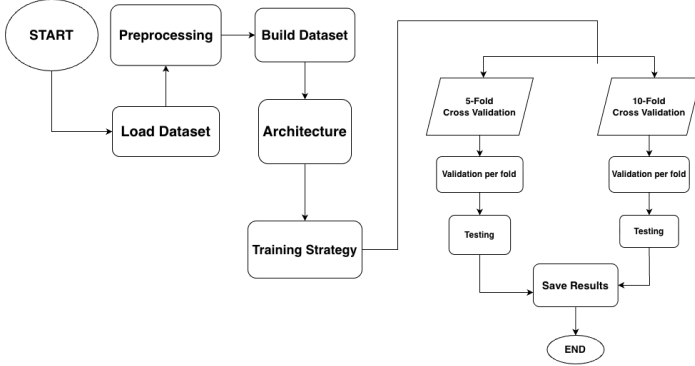


Fig. 1. This figure shows the overview of the whole pipeline

B. Preprocessing

In blind-sweep ultrasound acquisition, frames are collected as continuous sequences without requiring the operator to align standard anatomical planes. As illustrated in Fig. 2, the Obstetric Sweep Protocol (OSP) defines structured sweep trajectories, resulting in temporally correlated but spatially inconsistent data. In order to include short-time context, successive frames are stacked into groups of $S = 5$ slices (central frame: -2, +2), creating tensors of size (B, S, H, W) , which are then aggregated using channel-wise averaging before being passed to the encoder. Each frame is normalised by percentile-based clipping (1st - 99th percentile) and scaling to the range $[0,1]$. Images and masks are resized to a fixed resolution of 128×128 using bilinear interpolation of images and nearest-neighbor interpolation of masks. Annotations on the pixels are turned into binary masks and all preprocessing procedures are deterministic making them reproducible across experiments. These preprocessing steps help mitigate noise, motion artifacts, and variability introduced by blind-sweep acquisition while preserving relevant anatomical structures.

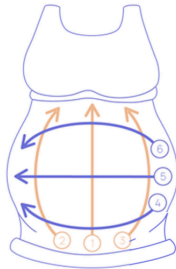


Fig. 2. The Obstetric Sweep Protocol (OSP). (a) OSP sweep trajectories. In orange, transverse sweeps (1–3); in violet, sagittal sweeps (4–6). Adapted from Sappia et al., Medical Image Analysis, 2025 (CC BY 4.0) [1].

C. Architecture

The network is an encoder-decoder (U-shaped) segmentation architecture implemented from scratch and adapted to accept a short temporal context. The input to the pipeline is sequence of frames of length $S = 5$ frames to restrict memory and calculate the five channels. The initial phase is the stacking of stack through channel-wise temporal averaging generating a representative frame in front of feature extraction. The overall U-shaped backbone follows standard downsampling - bottleneck - upsampling stages with skip connections. [2]

At multiple hierarchical stages in both encoder and decoder, the architecture incorporates an SSM-inspired gated large-receptive-field refinement block (denoted as Mamba2D). For an input tensor $X \in \mathbb{R}^{B \times C \times H \times W}$, the block computes a gating map using a 1×1 convolution, applies a grouped large-kernel convolution to capture broader spatial context, and projects the result through another 1×1 convolution. The final output is computed as

$$Y = \phi(\text{BN}(Z \odot G + X)), \quad (1)$$

where BN denotes Batch Normalization, ϕ represents the GELU activation, and $\odot$ indicates elementwise multiplication. The grouped large-kernel operation increases the effective receptive field while controlling computational cost, and the gating mechanism suppresses noisy responses while highlighting informative spatial regions. The block design is inspired by recent SSM based vision approaches and is integrated with the U-Net backbone for empirical evaluation [3].

Using $S = 5$ frames allows the model to capture short-range temporal context in ultrasound sweep sequences where adjacent frames contain correlated anatomical information. Channel-wise temporal averaging reduces speckle noise and frame-to-frame fluctuations while preserving consistent anatomical structures. Converting the stack into a single representative frame before the encoder reduces memory usage and computational overhead, which is particularly useful for medical datasets with limited training samples.

The architecture uses BatchNorm for stable training, GELU(Gaussian Error Linear Unit) activations for smooth nonlinear transformations, and MaxPool layers for progressive spatial downsampling. Decoder stages use transposed convolutions for upsampling and combine encoder features through skip connections. Training is performed using a segmentation loss with overlap-aware supervision, and model performance is evaluated using the Dice coefficient between predicted and ground-truth masks. Additionally, the predicted segmentation masks are used to estimate the fetal abdominal circumference through ellipse fitting, and the accuracy of the estimated circumference is evaluated using the NAE-AC.

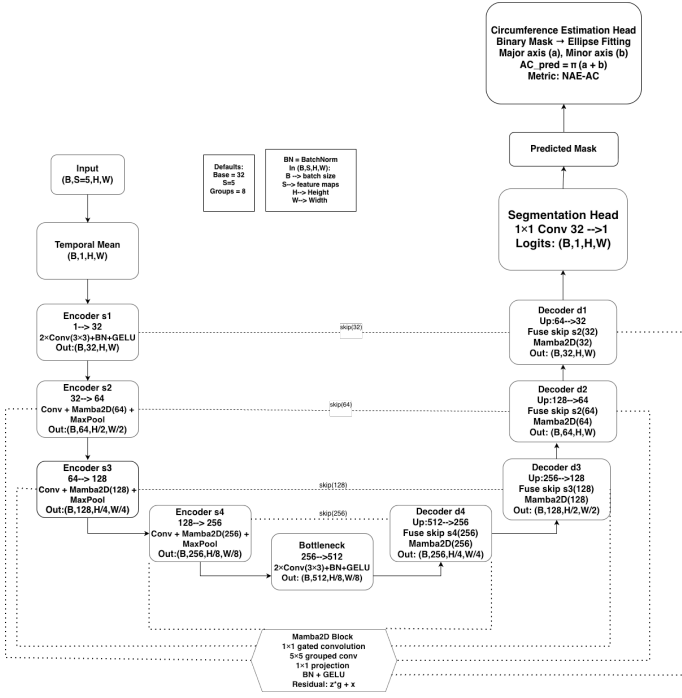


Fig. 3. A UNet-style encoder-decoder with Mamba2D gated grouped large-kernel refinement blocks at multiple scales; outputs segmentation and circumference logits [1,2].

III. EXPERIMENTAL SETUP AND RESULTS

A. Dataset

The experiments were conducted using the publicly available ACOUSLIC-AI training dataset containing prenatal ultrasound blind-sweep sequences for fetal abdominal segmentation [1]. The dataset consists of 2D B-mode ultrasound sweeps acquired by novice users in low-resource settings, which introduces challenges such as motion artifacts, variable image quality, and non-standard anatomical views [14], [20].

Only the public training set of 300 cases collected from public health units in Sierra Leone was used. Each case includes 6 ultrasound sweep sequences with pixel-level fetal abdomen annotations provided by trained readers [1]. Frames without visible fetal abdomen structures were removed during preprocessing to avoid irrelevant samples.

Before model input, frames were resized to a fixed resolution and intensity was normalized. Consecutive frames were grouped into short temporal stacks to provide contextual information from neighboring slices, improving robustness for ultrasound analysis tasks [14], [21]. The processed dataset was then divided into training, validation, and testing subsets following the experimental configuration used in the implementation. Additional details on data acquisition and annotation protocols are described in the ACOUSLIC-AI challenge report [1].

B. Training

The model was trained using the AdamW optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} .

All experiments used 2.5D input stacks of $S = 5$ frames resized to 128×128 . The batch size was set to 10 with 8 data loading workers, and dataset splitting was performed deterministically using a fixed random seed (42). Cases were split deterministically. After filtering empty frames, the annotated samples used in the reported run consisted of 216 training cases (4,753 annotated frames), 24 validation cases (609 annotated frames), and 60 test cases used as a holdout set (1,252 annotated frames). Two training strategies were evaluated to improve robustness: (1) five-fold cross-validation on the training pool, (2) ten-fold cross-validation using the same configuration. Performance was evaluated using the Dice coefficient for segmentation accuracy and NAE-AC to assess the accuracy of fetal abdominal circumference estimation. Across all experiments, the results were reported as average values.

C. Results

1) *five-fold cross-validation on the training pool*: Every fold was trained for 100 epochs using the same training configuration. Summary metrics obtained from this run are presented in Table I and Figure 4 shows representative segmentation results from the test set.

TABLE I
SUMMARY OF FIVE-FOLD CROSS-VALIDATION PERFORMANCE

Fold	Dice Score(Val set)	Dice Score(Test set)	NAE-AC(Test set)
Fold 1	0.8960	0.914	0.7569
Fold 2	0.9095	0.912	0.7572
Fold 3	0.9074	0.916	0.7575
Fold 4	0.9132	0.913	0.7583
Fold 5	0.9013	0.915	0.7571

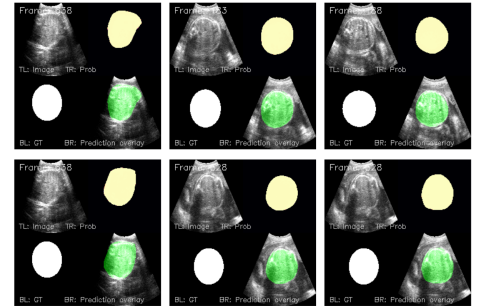


Fig. 4. Representative test-case segmentation results (TL: input frame; TR: predicted probability map; BL: ground-truth mask; BR: predicted segmentation overlay).

2) *ten-fold cross-validation on the training pool*: Every fold was trained for 100 epochs using the same training configuration. Summary metrics obtained from this run are presented in Table II. and Figure 5 shows representative segmentation results from the test set.

TABLE II
SUMMARY OF 10-FOLD CROSS-VALIDATION PERFORMANCE

Fold	Dice Score(Val set)	Dice Score(Test set)	NAE-AC(Test set)
Fold-1	0.866	0.906	0.7599
Fold-2	0.903	0.915	0.7576
Fold-3	0.898	0.922	0.7568
Fold-4	0.902	0.912	0.7573
Fold-5	0.899	0.905	0.7572
Fold-6	0.902	0.915	0.7565
Fold-7	0.898	0.916	0.7571
Fold-8	0.909	0.914	0.7561
Fold-9	0.912	0.912	0.7569
Fold-10	0.887	0.914	0.7594

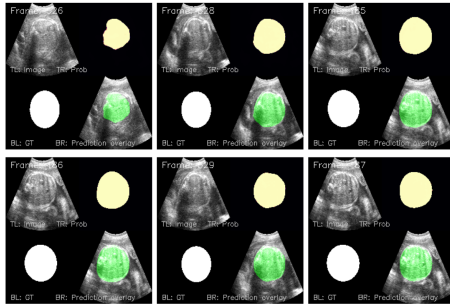


Fig. 5. Representative test-case segmentation (TL: input frame; TR: predicted probability map; BL: ground-truth mask; BR: predicted segmentation overlay).

IV. CONCLUSION

This study has discussed the use of a U-shaped encoder-decoder segmentation model and SSM-inspired large-receptive-field refinement blocks to segment fetal abdomen in ultrasound images. The experimental results show the average Dice Score 0.906 ± 0.006 and 0.897 ± 0.013 for five-fold and ten-fold cross-validation, respectively with corresponding NAE-AC values of 0.7574 ± 0.001 and 0.7572 ± 0.001 on ACOUSLIC-AI dataset. Such a framework can help determine the fetus growth using ultrasound imaging in low-resource settings.

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COMPLIANCE WITH ETHICAL STANDARDS

The preparation of this manuscript only involved the use of AI-assisted tools in grammar refinement and writing support. There were no AI-based data or experimental results that were utilized in this study and all experiments and analyses were performed by the authors themselves.

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